

George A. Schaible
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RESEARCH INTERESTS

Exploring the physiology, cellular biology, and ultrastructure of uncultured microorganisms using next-generation techniques to connect single-cell processes to ecosystem-scale impacts.

PROFESSIONAL APPOINTMENTS

— **University of California, Santa Barbara**

Molecular, Cellular, and Developmental Biology Department | June 2024 – present
Postdoctoral Researcher with Dr. Jean-Marie Volland

Research Focus: My research focuses on the sulfur-oxidizing bacterium *Thiovulum*, the fastest swimming bacterium known. My work achieved the first cultivation of *Thiovulum* on defined media since the organism's discovery over 240 years ago and combines genomics, expansion microscopy, and behavioral analysis to investigate how expanded intracytoplasmic respiratory membranes power ultrafast motility and emergent collective behavior at oxic-anoxic interfaces.

— **Montana State University**

Chemistry and Biochemistry Department | January 2024 – June 2024
Postdoctoral Researcher with Dr. Roland Hatzenpichler

Research Focus: I applied a cultivation-independent correlative microscopy workflow that links microbial identity, morphology, biochemistry, and physiology at the single-cell level to understanding microbial interactions in the human gut.

— **GlaxoSmithKline (GSK) Pharmaceuticals**

Microbiology Department | July 2016 – July 2018
Quality Control Technician

Focus: As a Quality Control Analyst in a GMP Grade C manufacturing environment, I ensured the quality and safety of vaccine products through daily analytical chemistry and microbiological testing of purified water, raw materials, and finished product in accordance with USP and FDA standards.

EDUCATION

— **Montana State University**

Molecular Bioscience Fellow | August 2018 – January 2024
PhD of Biochemistry

Research Focus: Under the advisement of Dr. Roland Hatzenpichler, my research focused on studying multicellular magnetotactic bacteria (MMB), the only currently known example of obligate multicellularity in bacteria. Because MMB have never been successfully cultivated, my research necessitated the use of next generation physiology techniques.

— **Montana State University**

Bozeman, MT | August 2010 – May 2014
Bachelor of Science in Microbial Biotechnology
Minors in Biochemistry and Astrobiology

Research Focus: Under the advisement of Dr. Dave Ward, I studied ecology and adaptation of the photosynthetic bacteria *Synechococcus* in Yellowstone National Park hot springs.

GRANTS AND FELLOWSHIPS

- **Harvey L. Karp Discovery Award (\$50,000)**
University of California, Santa Barbara | April 2026
- **Future Investigators in NASA Earth and Space Science and Technology (\$133,327)**
Montana State University | September 2020 – 2023
- **International Geobiology Course Fellow**
Summer course hosted by Caltech and funded by the Agouron Institute | 2022
- **Molecular Bioscience Fellow (fellowship covered stipend for 1st year of grad school)**
Montana State University | August 2018 – May 2023
- **Montana Academy of Science Student Research Grant (\$2,000)**
Montana State University | May 2019 – April 2020
- **Seed Grant to explore the use of Raman microspectroscopy (\$500)**
Montana State University | July 2019
- **Governors Energy Research Grant (\$1,500)**
Montana State University | Fall 2012 – Spring 2013
- **Schutter Family Scholarship**
Montana State University | Fall 2012 – Fall 2013
- **Undergraduate Scholars Program (two \$750 awards received)**
Montana State University | Fall 2012 – Spring 2013

AWARDS AND DISTINCTIONS

- **Best Postdoc Short Talk Award**
Joint Genome Institute NeLLi conference | 2025
- **PLOS Biology Editors Pick (<https://doi.org/10.1371/journal.pbio.3002638>)**
Editors pick for top microbiology and immunology paper | 2024
- **Cover image for Microbiology: an evolving science**
Textbook cover | 2023
- **Cover image of The ISME Journal**
Winner of the ISME journal cover competition | 2022

BROADER IMPACTS

- **Matters Microbial Podcast**
*<https://www.microbe.tv/mm/mm-076/> | January 2025
Guest on podcast where I shared the work performed during my PhD*
- **Founder and organizer of the Science Art Project**
*Montana State University | February 2021 – January 2024
The aim of this project is to bring science to the community through the medium of art*
- **Co-founder and president of the Astrobiology Journal Club**
*Montana State University | August 2019 – January 2024
Journal club aimed at creating a collaborative astrobiology community on campus*
- **Co-chair of Chemistry and Biochemistry Graduate Student Association**
*Montana State University | January 2020 – January 2022
Association of graduate students that provides a supportive environment for graduate students*
- **STEM Storyteller Fellow**
*Montana State University | August 2019 – May 2020
Multi-disciplinary fellowship program to improve science communication skills*

PEER REVIEWED PUBLICATIONS

- Borges, S. R., **Schaible, G. A.**, Sagasti, A. J., Teece, B. L., Soares, G. G., Barlow, E. V., Gangidine, A. (2024). Towards process-driven research: Stepping away from the binary biogenicity vs. abiogenicity approach. *Astrobiology*.
- **Schaible GA**, Cliff JB, Crandall JA, Bougoure JJ, Mathuri MN, Sessions AL, Atwood J, Hatzenpichler R. Comparing Raman and NanoSIMS for heavy water labeling of single cells. *Microbiology Spectrum*. 2025 May 30:e01659-24.
- **Schaible GA**, Jay ZJ, Cliff J, Schulz F, Gauvin C, Goudeau D, Malmstrom RR, Ruff SE, Edgcomb V, Hatzenpichler R. Multicellular magnetotactic bacteria are genetically heterogeneous consortia with metabolically differentiated cells. *PLoS Biology*. 2024 Jul 11;22(7):e3002638.
- Schaible MJ, Szeinbaum N, Bozdag GO, Chou L, Grefenstette N, Colón-Santos S, Rodriguez LE, Styczinski MJ, Thweatt JL, Todd ZR, ... **Schaible, G. A.**, ... Vázquez-Salazar A. Chapter 1: The Astrobiology Primer 3.0. *Astrobiology*. 2024 Mar 1;24(S1):S-4.
- Colón-Santos S, Vázquez-Salazar A, Adams A, Campillo-Balderas JA, Hernández-Morales R, Jácome R, Muñoz-Velasco I, Rodriguez LE, Schaible MJ, **Schaible GA**, Szeinbaum N. Chapter 2: What Is Life?. *Astrobiology*. 2024 Mar 1;24(S1):S-40.
- Bozdag GO, Szeinbaum N, Conlin PL, Chen K, Fos SM, Garcia A, Penev PI, **Schaible GA**, Trubl G. Chapter 5: Major Biological Innovations in the History of Life on Earth. *Astrobiology*. 2024 Mar 1;24(S1):S-107.
- Styczinski MJ, Glaser DM, Hooks M, Jia TZ, Johnson-Finn K, **Schaible GA**, Schaible MJ. Chapter 11: Astrobiology Education, Engagement, and Resources. *Astrobiology*. 2024 Mar 1;24(S1):S-216.
- **Schaible GA**, Kohtz AJ, Cliff J, Hatzenpichler R. Correlative SIP-FISH-Raman-SEM-NanoSIMS links identity, morphology, biochemistry, and physiology of environmental microbes. *ISME communications*. 2022 Dec;2(1):52.
- Nowack S, Olsen MT, **Schaible GA**, Becraft ED, Shen G, Klapper I, Bryant DA, Ward DM. The molecular dimension of microbial species: 2. Synechococcus strains representative of putative ecotypes inhabiting different depths in the Mushroom Spring microbial mat exhibit different adaptive and acclimative responses to light. *Frontiers in Microbiology*. 2015 Jun 29;6:626.
- **Schaible GA**, Strobel GA, Mends MT, Geary B, Sears J. Characterization of an endophytic Gloeosporium sp. and its novel bioactivity with “synergists”. *Microbial ecology*. 2015 Jul;70:41-50.
- Strobel G, Booth E, **Schaible GA**, Mends MT, Sears J, Geary B. The Paleobiosphere: a novel device for the in vivo testing of hydrocarbon producing-utilizing microorganisms. *Biotechnology letters*. 2013 Apr;35:539-52.

MANUSCRIPTS IN REVIEW, SUBMITTED, OR IN PREPARATION

- **Schaible GA**, Woyke T, Brauns F, Contarini PE, Natesan D, Beahrs E, Szabo G, Elise JP, Gros O, Volland JM. The fastest known bacterium powers its motility with mitochondria-like respiratory membranes In-prep
- **Schaible GA**, Melendrez MC, Olsen M, Becraft ED, Nowack S, Wood JM, Bryant DA, Cohan FM, and Ward DM. Genomic and Phenotypic Insights into Light Adaptations of A-like Synechococcus Species. In-prep
- Malas, M., Rodriguez-Colon, B. J., Wallet, E., **Schaible, G. A.**, Brassell, S. C., Silverman, S., Betts, M. N., Magyar, J. S., Webb, S., Sessions, A. L., Orphan, V. J., Goffredi, S., Shapiro R. S. (2024). Temporal Diversity within Cretaceous Methane Seeps, Great Valley Group, Central California. In-prep

CONFERENCES AND INVITED TALKS

- **California State University, Fullerton**
Talk | “The world's fastest bacterium: ultrafast motility and emergent collective behavior in *Thiovulum*” | April 2026
- **New Lineages of Life (NeLLi)**
Talk | “Cultivation of *Thiovulum* reveals record-breaking motility and emergent collective behavior” | November 2025
- **Applied and Environmental Microbiology Gordon Research Conference (AEM GRC)**
Talk & Poster | “Multicellular bacterial consortia are metabolically differentiated and not clonal” | July 2023
- **Astrobiology Graduate Conference (AbGradCon)**
Talk & Poster | “Multicellular bacterial consortia are metabolically differentiated and not clonal” | May 2023
- **International Society for Microbial Ecology (ISME)**
Talk & Poster | “Diversity, morphology, physiology, and division of labor of obligate multicellular bacteria” | August 2022
- **Montana Nanotechnology**
Poster | “Correlative Raman-FM-SEM-NanoSIMS links identity, biochemistry, and morphology of environmental microbes” | October 2021
- **New Lineages of Life (NeLLi)**
Talk | “Correlative Raman-FM-SEM-NanoSIMS links identity, biochemistry, and morphology of environmental microbes” | October 2021
- **Astrobiology Graduate Conference (AbGradCon)**
Talk | “Cellular Differentiation Within Multicellular Magnetotactic Bacteria: Implications to the Evolution of Complex Life on Earth” | September 2021
- **Woods Hole Oceanographic Institute (WHOI)**
Talk | “Exploring the evolution of bacterial complexity using multicellular magnetotactic bacteria” | September 2021
- **American Society of Microbiology (ASM)**
Poster | “MMB: Organized Complexity in the Domain Bacteria” | June 2021
- **American Association for the Advancement of Science (AAAS)**
Poster | “MMB: Organized Complexity in the Domain Bacteria” | February 2020

TECHNICAL SKILLS

- **Laboratory techniques**
Experimental design and execution | *organization and cleanliness* | *aseptic techniques* | *pipetting and liquid handling* | *DNA extraction* | *PCR (polymerase chain reaction)* | *high molecular weight DNA extraction* | *genomic DNA library construction* | *Nanopore sequencing* | *reagent and media preparation* | *cultivation of lab strains and environmental microbes* | *cell line maintenance* | *enzymatic assays* | *catalyzed reporter deposition fluorescence in situ hybridization (CARD-FISH)* | *double labeling of oligonucleotide probes (DOPE)-FISH* | *Clone-FISH* | *substrate analog probing (SAP)* | *bioorthogonal non-canonical amino acid tagging (BONCAT)* | *click chemistry* | *stable isotope probing (SIP)* | *fluorescence activated cell sorting (FACS)* | *spectrophotometry* | *preparation of laboratory protocols* | *safety and quality culture*

TECHNICAL SKILLS *CONTINUED*

— **Microscopy, spectroscopy, and spectrometry techniques**

Brightfield microscopy | fluorescence microscopy | laser microdissection microscopy | confocal laser scanning microscopy | confocal Raman microspectroscopy | stimulated Raman spectroscopy (SRS) | gas chromatography and mass spectrometry (GC/MS) | nano-scale secondary ion mass spectrometry (NanoSIMS) | scanning electron microscopy (SEM) | backscatter electron microscopy | energy-dispersive X-ray spectroscopy (EDS) | cathode luminescence | transmission electron microscopy (TEM) | expansion microscopy (Magnify and U-ExM) | correlative microscopy

— **Fieldwork**

Planning and preparation for multiple day sampling trips | diel sampling experience | sampling of hot springs, salt marsh, alkaline lake, plant material, and rock fossils | rhizon pore water collection | sulfide assay | ferrozine assay | sample site description and mapping

— **Bioinformatics techniques**

Command line | analysis of short read Illumina sequencing | FISH probe design | phylogenetic analysis | genome annotation | metagenomic analysis | JGI IMG | large metadata set management

— **Data analysis**

Proficiency in R for statistical analysis and data visualization | experience with Python | ability to write and organize multiple coding languages for publication on Github

— **Engineering experience**

Autocad design | 3D printing | photolithography | preparation of microfluidic PDMS | bonding of microfluidic devices | use of syringe pumps | soldering and repair of lab equipment

RELATIVE WORK EXPERIENCE

— **GlaxoSmithKline (GSK) Pharmaceuticals**

Quality Control Technician | Hamilton, MT | July 2016 – July 2018

Supervisor: Tate Simmons | tate.simmons@SRGwoolf.com

I worked in a full Good Manufacturing Practice (GMP) Grade C environment ensuring that our product was made in concordance with United States Pharmacopeia (USP) and Food and Drug Administration (FDA) standards. This involved daily testing of purified water, product, and raw materials using analytical chemistry and microbiological approaches. I was required to keep accurate and contemporaneous records during any and all GMP activity. Daily interactive meetings were held where I would coordinate and work closely with fellow employees. In this position I employed the use of the 5S standard to streamline work and efficiency as a means to lower spending as well as to mitigate deviations of our procedures.

— **CIEI English Immersion Program**

Teacher | Beijing, China | September 2015 – January 2016

Supervisor: Leslie Dong | cieieipbj@yahoo.com

I worked as an English teacher at public schools in Beijing, China. My job required me to make daily lesson plans for 6th, 5th, and 2nd grade students, each grade requiring topics that were specific to their level of English. I would teach twenty-five classes a week, each with more than forty students, thus requiring me to teach over one thousand Chinese students within a single week. I was responsible for managing the classes to ensure students behaved as I interacted with them to create a fun and conducive learning environment.

RELATIVE WORK EXPERIENCE *CONTINUED*

— **Montana State University**

Laboratory Technician III | Bozeman, MT | June 2012 – August 2015

Supervisor: Dr. David Ward | umdbw@montana.edu

I was responsible for the upkeep of the lab as well as assisting in several *in vitro* projects. My daily tasks included the ordering of laboratory supplies, mentorship to undergraduates, as well as performing cell culturing, DNA extraction and PCR, data analysis, flow cytometry, fluorescence microscopy, cryogenic preservation of bacteria, and occasional field work at our research sites at hot springs in Yellowstone National Park.

— **Montana State University**

Research Assistant | Bozeman, MT | November 2010 – December 2013

Supervisor: Dr. Gary Strobel | uplgs@montana.edu

During this research position, I was tasked with the isolation and characterization of fungal endophytes. The endophytes we studied were found growing in plant material sourced from foreign tropical rainforests and were studied for medical and biotechnological applications. My primary responsibilities included the employment of molecular techniques, such as DNA isolation and PCR, cultivation and culturing methods, analysis of gas chromatography and mass spectroscopy (GCMS) data, as well as general laboratory upkeep.